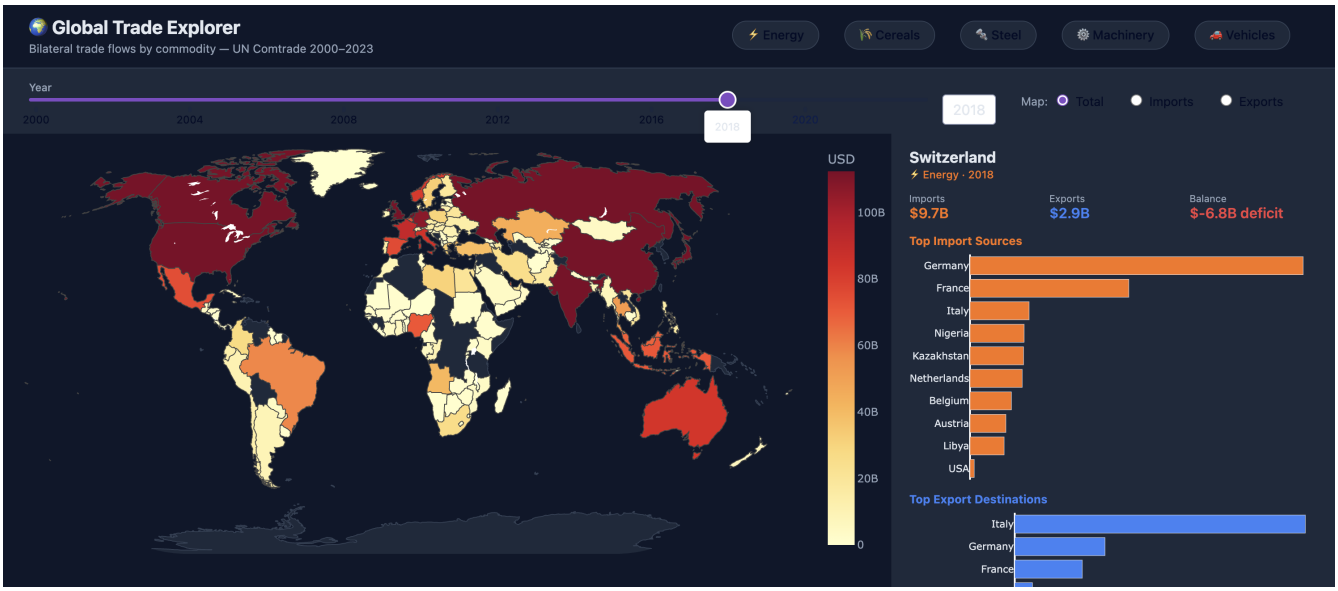
# Global Trade Explorer

Dependencies, Flows & Disruptions

COM-480 Data Visualization • Milestone 2 • Team Vizion • April 17, 2026

**Live prototype:** https://global-trade-viz.onrender.com **Repo:** https://github.com/com-480-data-visualizatio
Vizion

*Render free tier — first visit after inactivity may cold-start for ∼30–60 s while the container reloads; subsequent interactions are instant.*




Global Trade Explorer - choropleth world map with commodity selector, year slider, and click-to-drill country panel. Example: Switzerland, energy, 2018.

## 1 Project Goal & Audience

Global trade is the invisible scaffolding of daily life: the wheat in bread, the steel in buildings, the gas that heats homes. Yet most people - and most existing trade dashboards - stop at *totals*: how much was traded, by whom, in what year. That framing hides the most important questions of the 2020s: *how concentrated is each country's supply? How exposed are we to a single partner? What happens when a corridor breaks?* The 2022 energy shock, the 2020 semiconductor shortage, and the Red Sea disruptions all made these structural questions front-page news.

Our goal is to build an **interactive web dashboard that makes the structure of global trade legible**, not just its volume. Using UN Comtrade bilateral flows from 2000–2023, it lets a user choose a commodity family - **energy** (HS 27), **cereals** (HS 10), **iron & steel** (HS 72), **machinery** (HS 84), or **vehicles** (HS 87) - and answer three questions: (1) *Who trades what with whom?* (the choropleth and Sankey), (2) *How dependent is each country, and on whom?* (top-partner share, HHI, diversification over time), (3) *How has this structure shifted, and what would happen if it were disrupted?* (top movers, the dependency race, and the stretch-goal disruption simulator).

Existing platforms (OEC, World Bank, UN Comtrade dashboards) answer (1) well and (2) only partially; none address (3) interactively. Our contribution is pairing a familiar choropleth entry point with *progressively deeper analytical views* - time series with shock bands, Sankey flows, a per-country dependency panel, top movers, an animated dependency race - and layering on a *what-if* disruption mode on top.

**Audience.** Students, researchers, journalists, and engaged citizens interested in global economics, geopolitics, and supply-chain resilience. The design prioritizes *progressive disclosure*: a newcomer can read the map; a power user can drill into HHI values, animate the dependency race, and compare two countries side-by-side.

## 2 Dataset

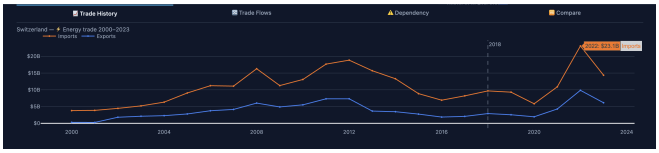
UN Comtrade, annual bilateral flows (reporter × partner × HS code × flow × USD), 2000–2023. After cleaning (year filter, ISO-3 normalization, removal of aggregate entities such as "World" and "Areas, nes") we ship three processed commodities (energy 393 k rows, cereals 230 k, steel 329 k) and are downloading two more (machinery, vehicles) under the Comtrade free-tier quota. Known limitations: re-export inflation for hub countries (NLD, BEL, SGP), nominal USD (no inflation adjustment), sparse coverage pre-2005.

## 3 Visualizations

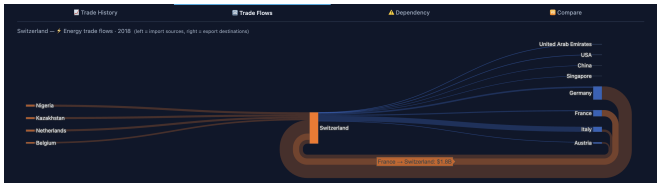
All views below are either working in the prototype or scaffolded behind a tab. Screenshots are taken directly from the running dashboard.

### 3.1 (A) Choropleth World Map - entry point

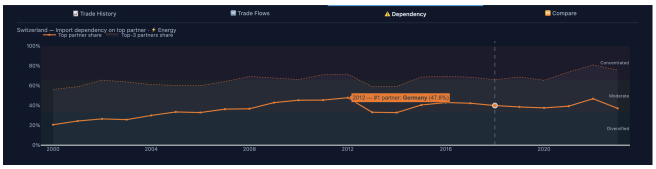
*Goal: give every user a first answer to "who trades how much?" in one glance, and route them to everything else.* Log-scaled USD, year slider (2000–2023), commodity dropdown, and an Import/Export/Total toggle. Clicking a country opens the detail panel (B–E). The log scale is deliberate: raw USD totals span five orders of magnitude, so a linear scale would flatten everything outside the top-10. See the hero screenshot on page 1.




**(a) (B) Trade History.** *Goal: expose shocks and long-run trends for the selected country.* Dual import/export lines across 2000–2023 with the active year marked; planned shock bands (2008, 2014–16, 2020, 2022) contextualize dips and spikes.

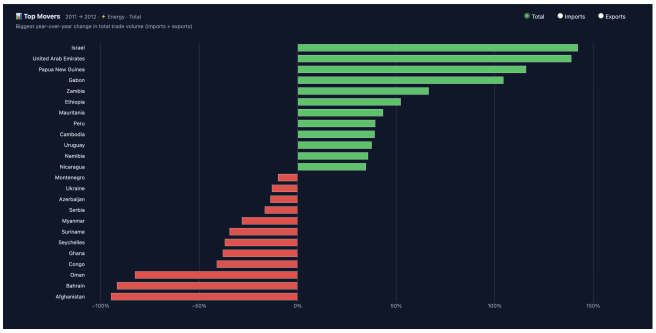



**(b) (C) Sankey Flow.** *Goal: show the bilateral structure of a country's trade - who it buys from and sells to, and in what proportion.* Ribbon width $\propto$ USD; left = import sources, right = export destinations.

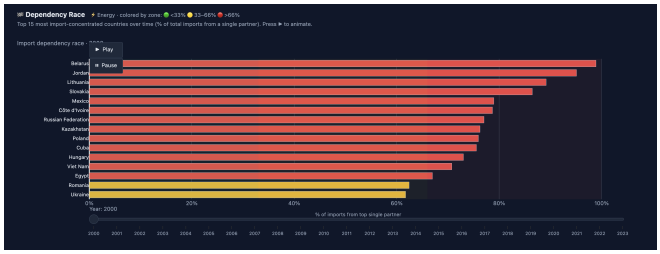



**(a) (D) Dependency.** *Goal: quantify concentration risk.* Top-partner and top-3-partners share plotted over time with diversified / moderate / concentrated bands, so the user sees at a glance whether a country is locked into one supplier or has diversified.

**(b) (G) Compare.** *Goal: answer "how does country X differ from country Y?"* Totals, overlaid history, and overlaid dependency curves for two countries in the same view, useful for peer comparisons (e.g. Germany vs. Japan on steel).




**(a) (E) Top Movers.** *Goal: surface the biggest year-over-year shifts.* Diverging bars (green gainers / red losers) of % change vs. the previous year; a trade-volume floor filters out low-volume noise so rankings reflect genuine structural moves.




**(b) (F) Dependency Race.** *Goal: reveal how concentration evolves.* Animated ranking of "% of imports from top single partner" over 2000–2023; plays back decades of diversification or lock-in in a few seconds.

# 4 MVP vs Extensions

**Core (MVP - must ship for milestone 3).** Each piece is independent: (1)–(3) rely only on `country_summary.csv`; (4)–(5) rely only on `partner_flow.csv`. They share no state other than the current selection, so any ordering ships.

1. Choropleth map + year slider + import/export toggle
2. Click-to-drill country detail panel (top-10 partners + history)
3. Commodity dropdown covering $\geq 3$ commodities
4. Sankey diagram of top bilateral flows
5. Dependency bar (top-5 partners + HHI) per country
6. Public deployment (*done* - see live link above)

**Extensions (droppable without breaking the narrative).**

- **Disruption simulator** - remove a country/link; recompute who loses what share of supply (propagation via re-routing).
- **Chord diagram** alternative view of the network (D3 embed).

- **Top Movers** and **Dependency Race** (already scaffolded).
- **"Explain this chart"** narrative side-captions.
- **Search-as-you-type** country finder.
- **Mobile-friendly** responsive layout.

# 5 Tools & Lectures

| Need | Tool | Lecture(s) |
|---|---|---|
| Map projection | Plotly `choropleth`, ISO-3 codes | Maps; Perception |
| Time series | Plotly lines + shock bands | Time-series; Annotation |
| Flows | Plotly `sankey` | Graphs & networks |
| Rankings / composition | Plotly `bar` | Rankings; Composition |
| Animation | Plotly `frames` | Animation; Interaction |
| Layout & callbacks | **Dash** (Python, Flask backend) | Interaction; Web viz tooling |
| Data wrangling | **pandas**, **NumPy** | (prerequisite) |
| Deployment | Render free tier, GitHub auto-deploy | (engineering) |

Future lectures we expect to lean on: **Storytelling** (narrative captions), **Graphs & Networks** (Sankey refinement, optional force layout), **Evaluation & Perception** (color scales, legend validation).

# 6 Prototype Status & Risks

**Running in production** at global-trade-viz.onrender.com: choropleth, year slider, commodity dropdown, country detail panel with Trade History / Sankey / Dependency / Compare tabs, global Top Movers chart, Dependency Race animation, preprocessing pipeline, EDA notebook.

**Gaps before milestone 3.** Finish machinery & vehicles downloads; ship the disruption simulator; polish color scale and mobile layout; tune cold-start.

**Risks.** *API quota* (500 calls/day) - mitigated by per-commodity CSV caching. *Missing ISO-3 codes* ($\sim$10% of partners) - manual override in `src/utils.py`. *Re-export inflation* - flagged in the EDA; dependency metric uses direct partner share. *Scope creep* - the MVP (§4) is independent of every extension, so extensions can be dropped without breaking the story.